Schematic auto-placer: before and after

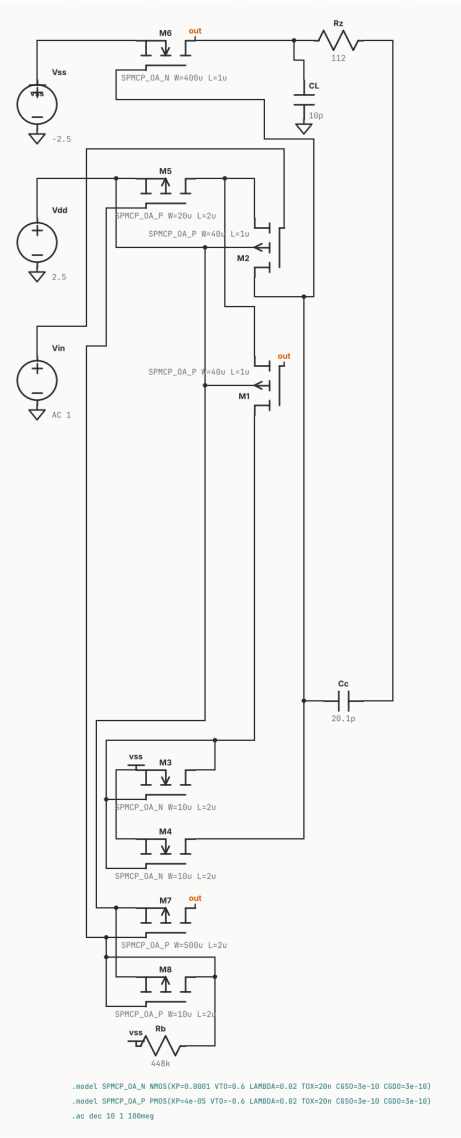
Readability score 0 to 100 (higher is easier to read). Every drawing, before and after, is checked pin by pin to be the netlist's circuit.

circuit	before	after	crossings	seconds	blocks found
LDO regulator (template)	76	100	2 -> 0	1.03 -> 1.4	-
Buck converter (template)	69	88	2 -> 1	0.65 -> 2.4	-
Bandgap reference (template)	59	95	3 -> 0	0.75 -> 2.4	-
Boost converter, closed loop (template)	43	82	4 -> 1	0.9 -> 3.9	-
Flyback converter, closed loop (template)	refused	81	- -> 1	0.0 -> 5.4	transformer
Band-pass filter (template)	88	88	1 -> 1	0.21 -> 0.2	-
Instrumentation amplifier (no template)	26	75	5 -> 2	0.87 -> 1.9	-
8-MOSFET op-amp, transistor level (no template)	4	51	6 -> 4	1.49 -> 5.1	diff_pair, mirror
State-variable filter, 3 op-amps (no template)	33	86	5 -> 1	0.63 -> 2.3	-
BJT diff pair, mirror load, follower (no template)	67	100	2 -> 0	1.1 -> 1.9	diff_pair, mirror
Wien-bridge oscillator (no template)	96	93	0 -> 0	0.33 -> 0.8	-
Class-AB power stage with op-amp (no template)	0	88	8 -> 1	3.75 -> 2.2	push_pull
4th-order Sallen-Key low-pass (no template)	100	100	0 -> 0	0.34 -> 0.3	-
Bare flyback with transformer (no template)	refused	98	- -> 0	0.0 -> 1.4	transformer

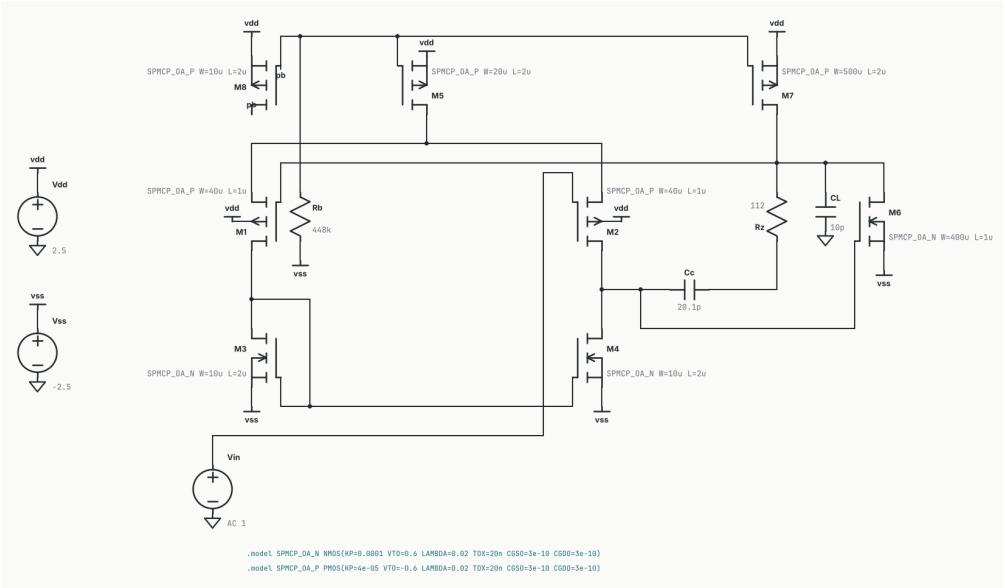
Average after: 88. Average before: 55 over the 12 it could draw (it refused the 2 with a transformer).
What changed: familiar blocks (diff pair, current mirror, push-pull, transformer) are recognised from the wiring and drawn the familiar way; placement tries a few dozen layouts and keeps the best correct one; busy supplies can be rail labels; the score also counts label jumps and page shape. The score still cannot judge everything a person sees, so the pictures follow.

8-MOSFET op-amp, transistor level (no template): 4 -> 51

before (1.0.0)

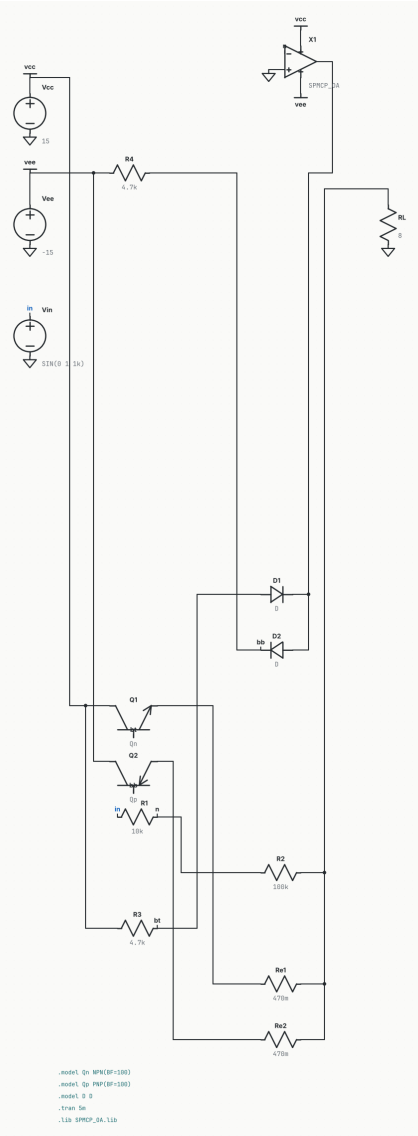


after (1.1.0)

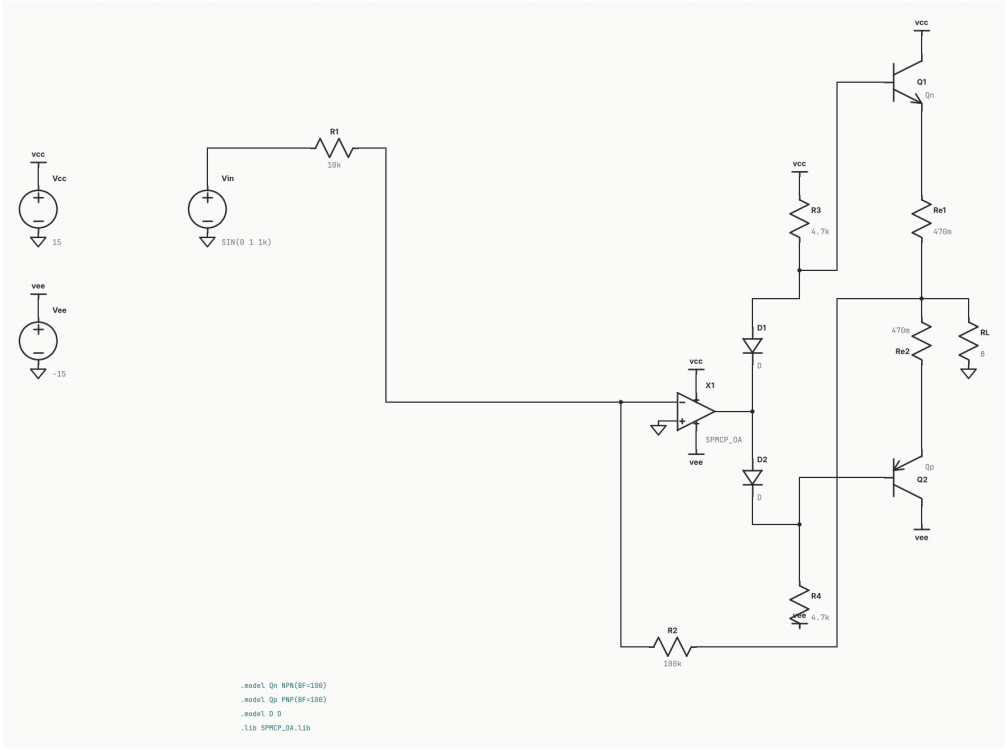


Class-AB power stage with op-amp (no template): 0 -> 88

before (1.0.0)

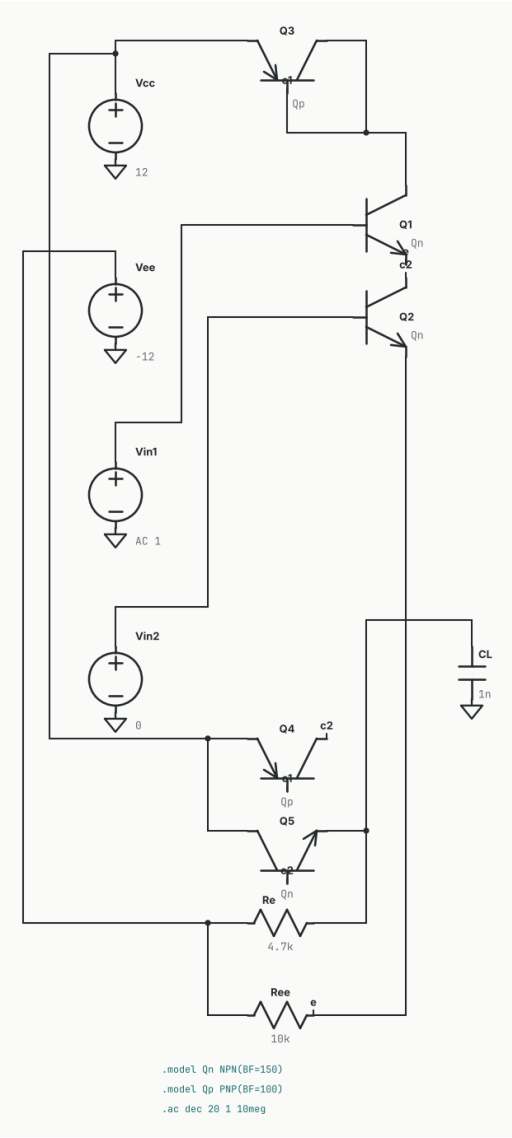


after (1.1.0)

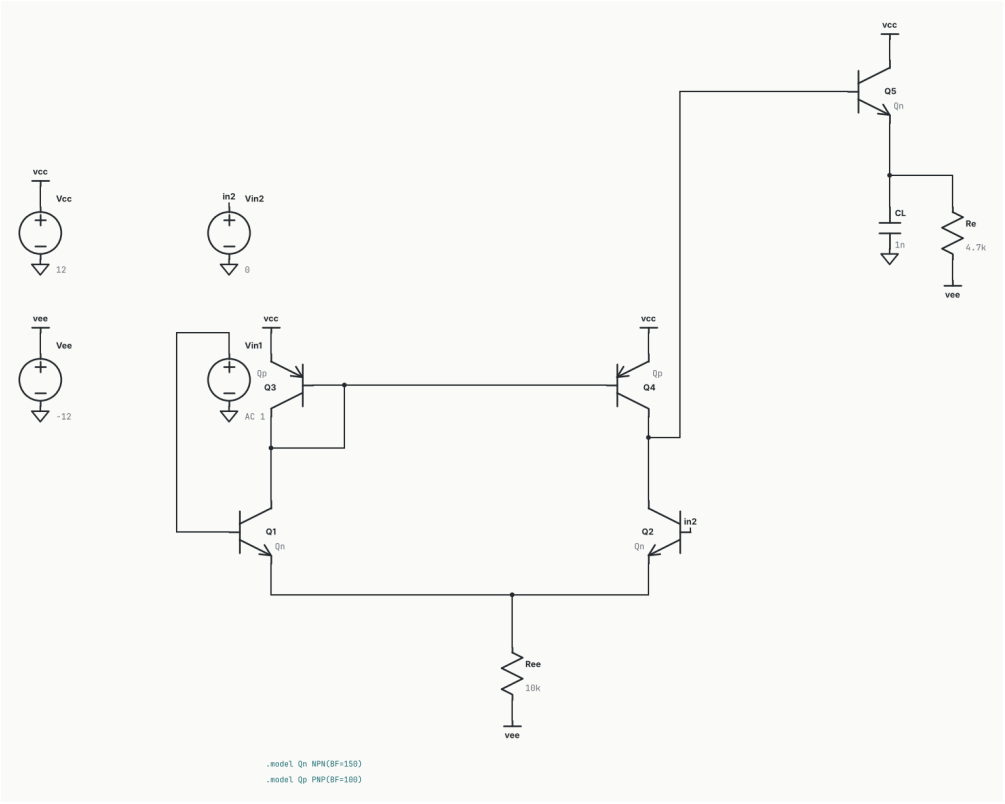


BJT diff pair, mirror load, follower (no template): 67 -> 100

before (1.0.0)

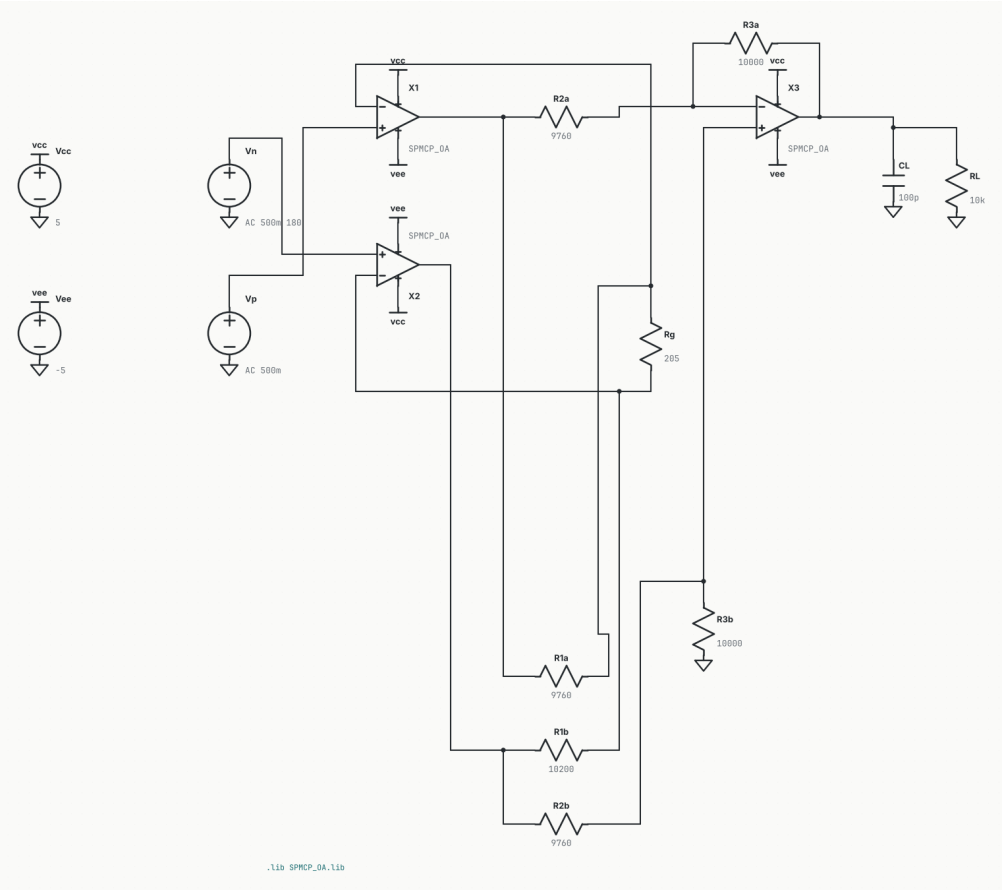


after (1.1.0)

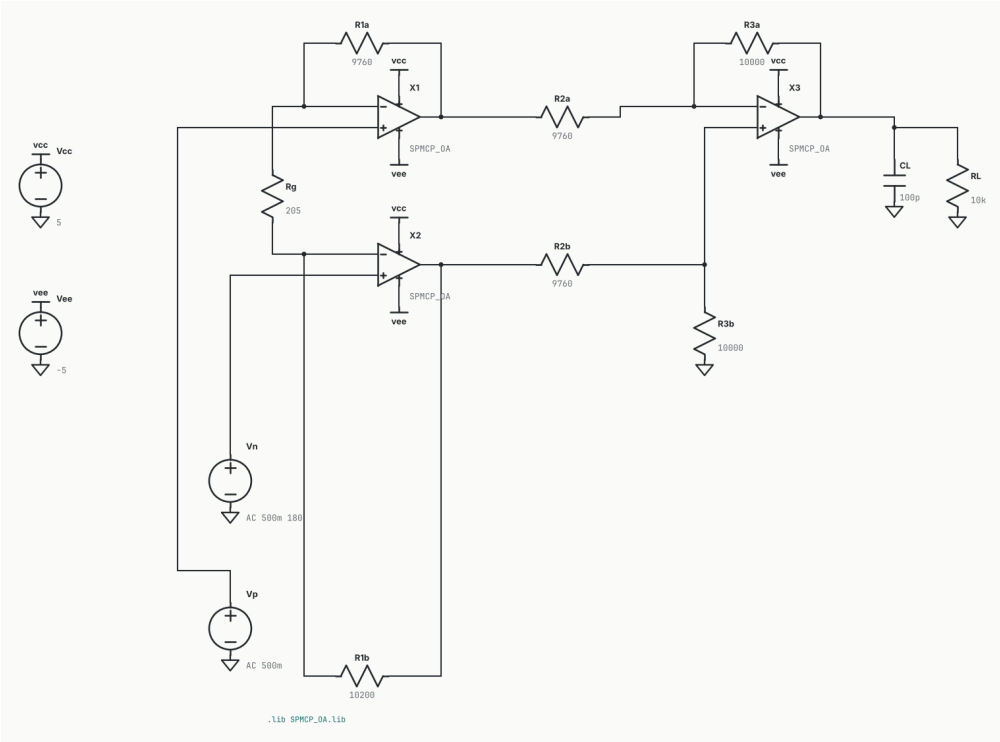


Instrumentation amplifier (no template): 26 -> 75

before (1.0.0)



after (1.1.0)

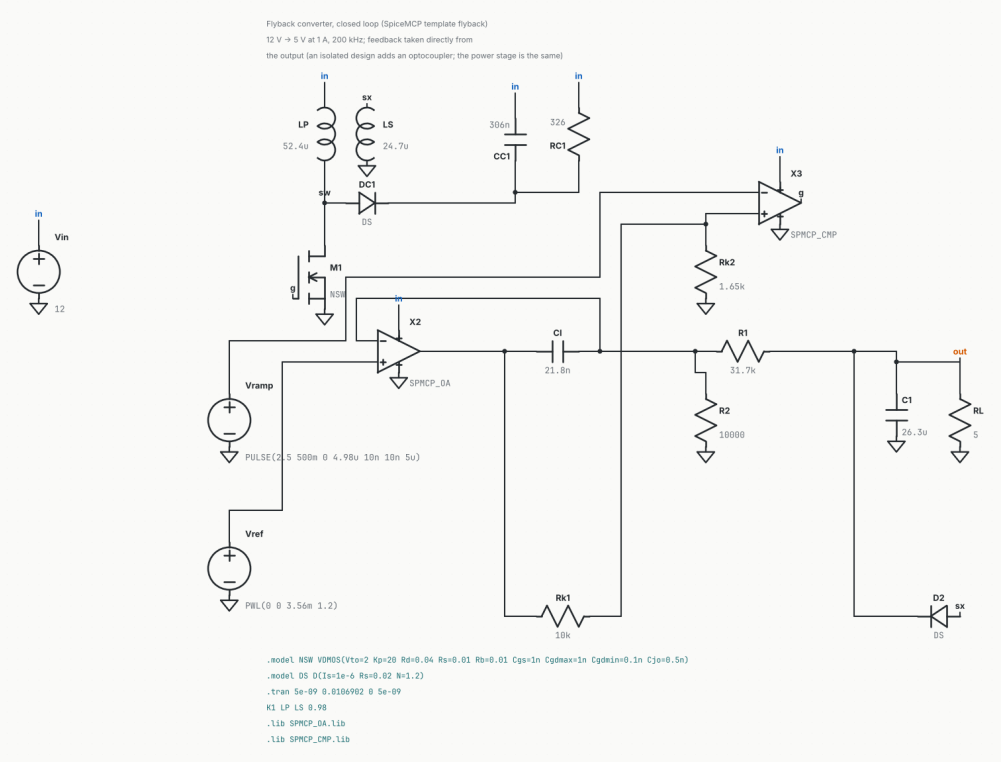


Flyback converter, closed loop (template): refused -> 81

before (1.0.0)

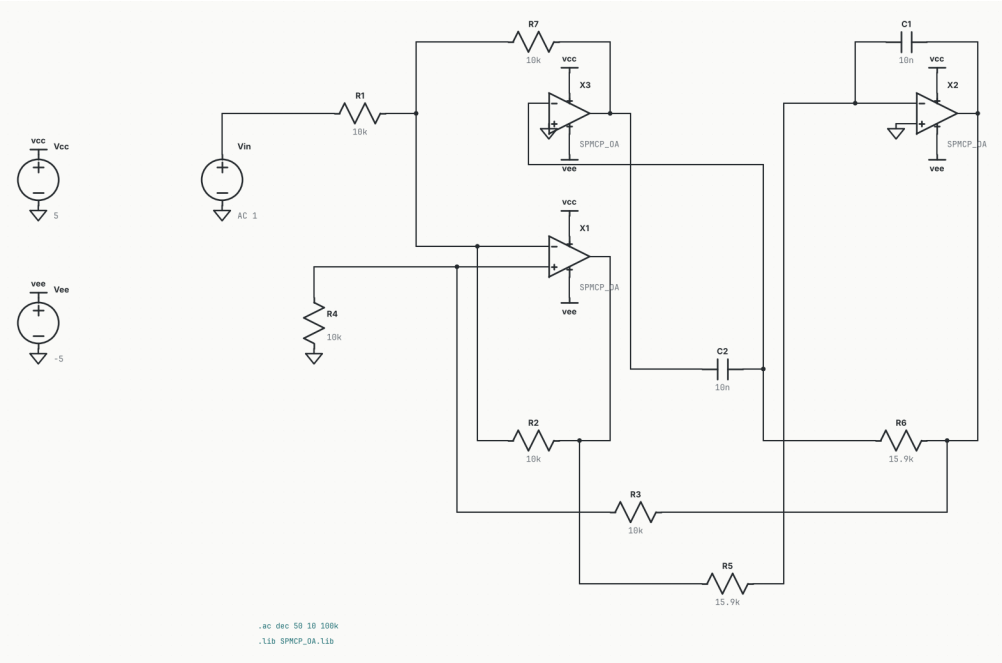
not drawn: refused: coupled windings (K) not supported

after (1.1.0)

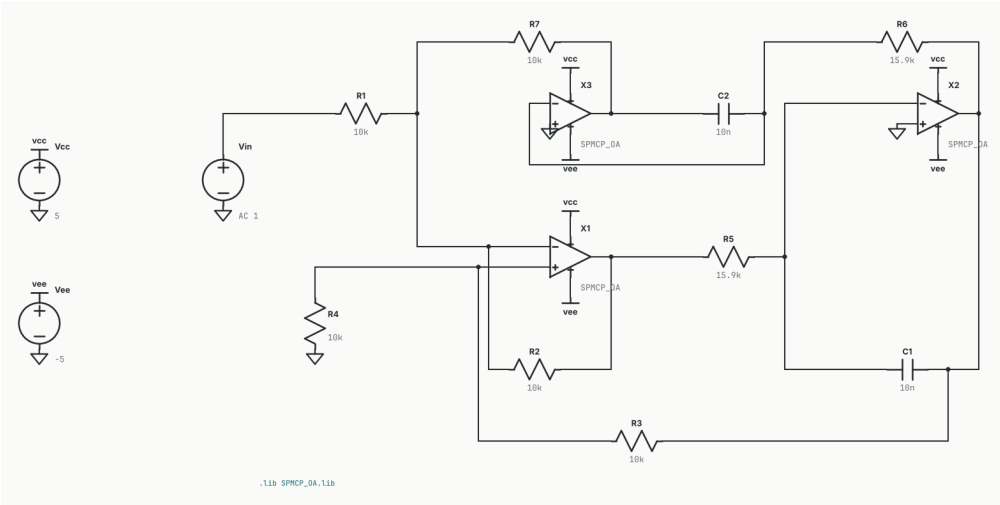


State-variable filter, 3 op-amps (no template): 33 -> 86

before (1.0.0)

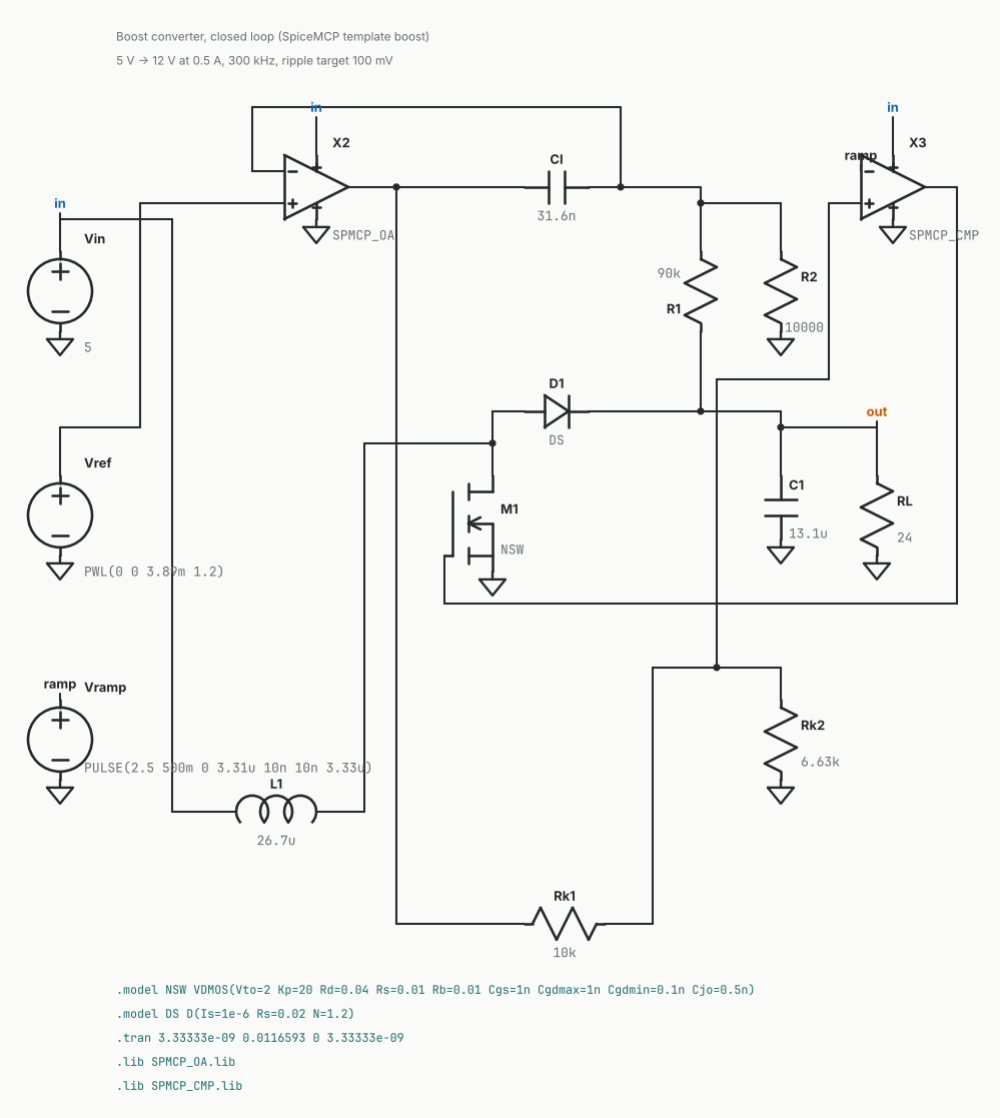


after (1.1.0)

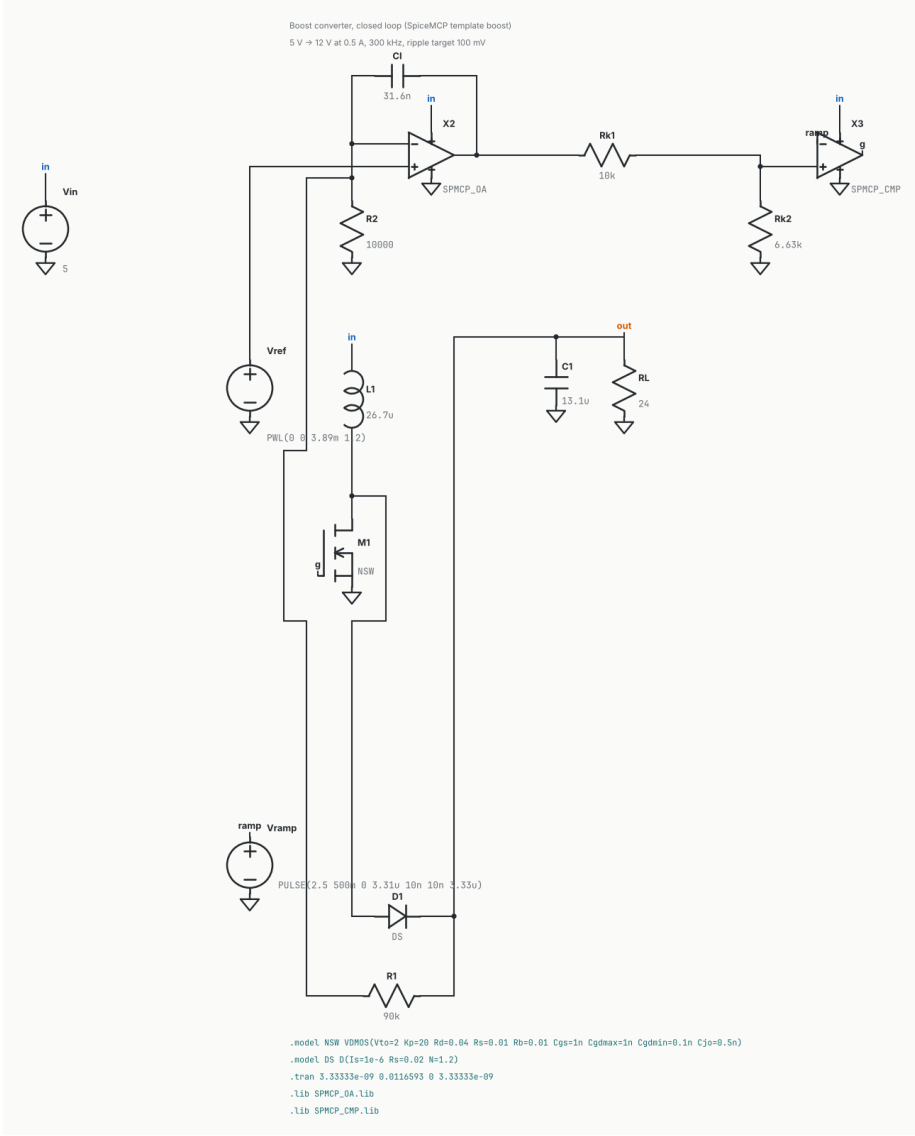


Boost converter, closed loop (template): 43 -> 82

before (1.0.0)

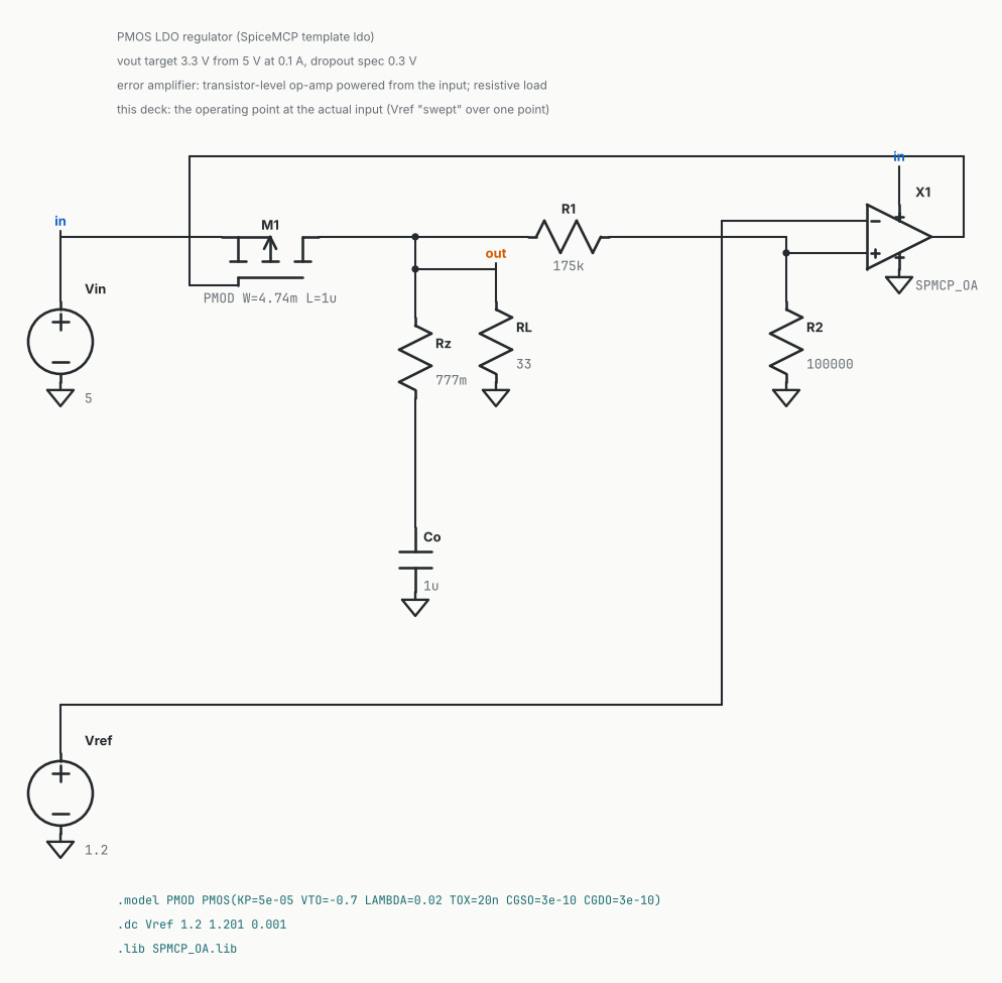


after (1.1.0)



LDO regulator (template): 76 -> 100

before (1.0.0)



after (1.1.0)

